

Quality Assurance Engineering Intern – Assessment

SMS Credit Reload Feature (MikeTango / GoSMS) · Syed Muhammad Aziz Ghaus · azizghaus14@gmail.com · +60 17-414 7089

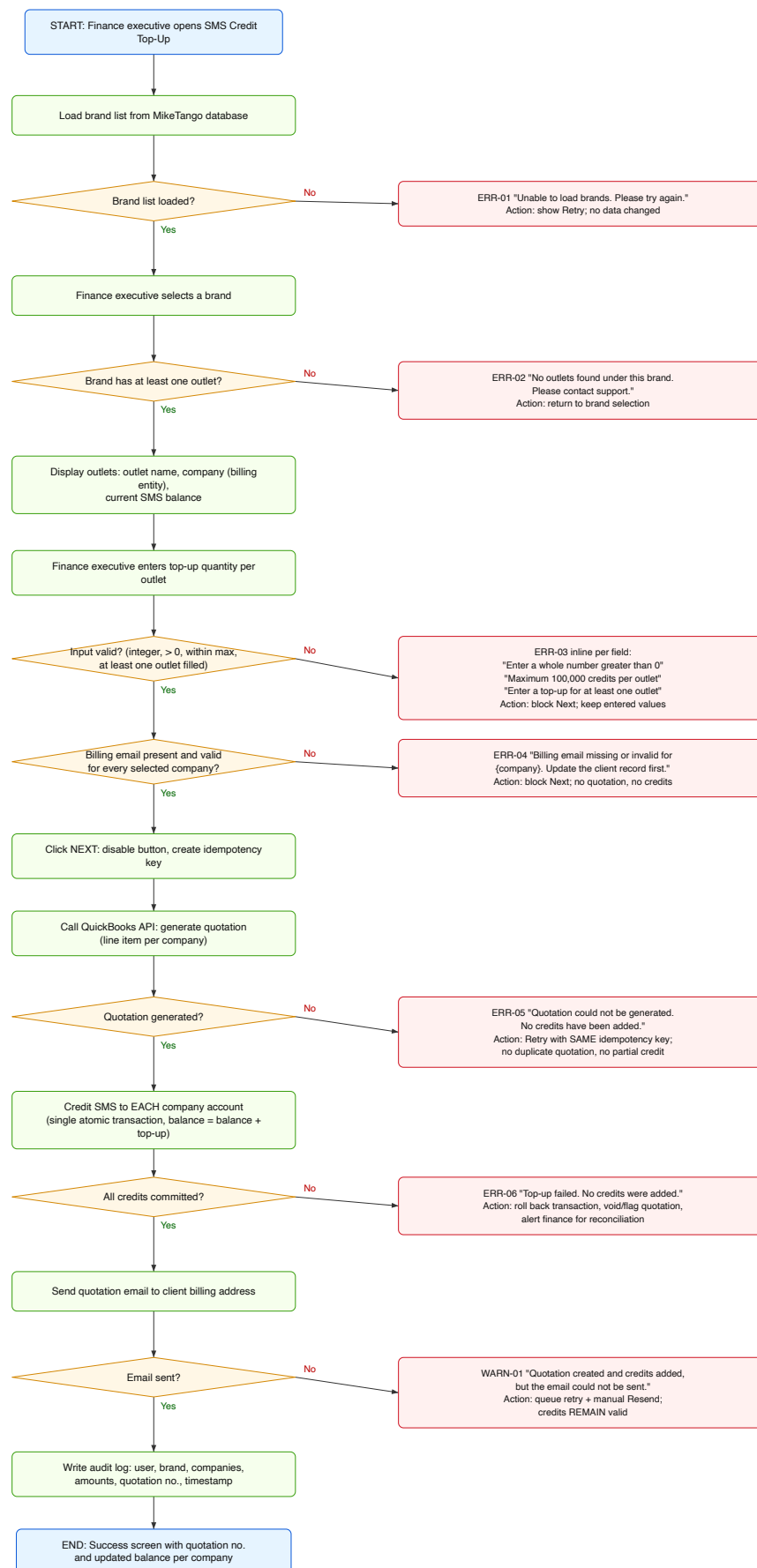
1. Assumptions

- SMS credits are held per **company (billing entity)**, not per brand. A brand may span several companies, so one submission can produce charges to more than one billing entity.
- The commit point is the **quotation**: credits are only added once QuickBooks confirms the quotation was generated.
- Emailing the quotation is treated as a **separate, non-blocking step**. If the email fails, the quotation and the credits stay valid and the email is retried, because reversing credits that the client has already been charged for would be worse than a delayed email.
- Crediting across several companies happens in a **single atomic transaction**, so a partial top-up can never occur.
- Each submission carries an **idempotency key** so a retry or double click cannot create a second quotation or double credits.

2. State model

State	Meaning
S0	Page open, no brand selected
S1	Brand selected, outlets loaded
S2	Valid top-up amounts entered
S3	Submitting (quotation request in flight)
S4	Quotation generated
S5	Credits committed
S6	Complete (emailed, audit logged, success screen)
E1–E7	Error states, one per failure point in the flowchart

3. Flowchart (including error handling and messages)



4. Test cases

Every state transition in the flowchart is covered exactly once. TC-06 covers a single transition (valid → validation error) and is written data-driven rather than repeated as six near-identical cases. TC-21 to TC-26 cover business rules that the state machine alone does not express.

ID	Tr.	Title	Precondition	Steps	Expected result	Type
TC-01	T01	Select a brand that has outlets	Brand list loaded; brand 'SilverTree' has 3 outlets across 2 companies	Select 'SilverTree'	Outlet list shows all 3 outlets with outlet name, company (billing entity) and current SMS balance	Positive
TC-02	T02	Brand list fails to load	Database/API for brand list unavailable	Open the Top-Up page	ERR-01 shown with Retry. No brand list, no further action possible. Nothing written to any account	Negative
TC-03	T03	Retry after brand list failure	ERR-01 displayed; service restored	Click Retry	Brand list loads; page returns to brand selection state	Recovery
TC-04	T04	Brand with no outlets	Brand 'EmptyBrand' exists with 0 active outlets	Select 'EmptyBrand'	ERR-02 shown. Next is disabled. User returned to brand selection	Negative
TC-05	T05	Enter valid top-up quantities	Outlets displayed	Enter 500 for outlet A, 1000 for outlet B, leave outlet C blank	Values accepted, no validation errors, Next becomes enabled	Positive
TC-06	T06	Invalid top-up input (data-driven, one transition)	Outlets displayed	Enter each in turn: 0 / -50 / 1.5 / abc / blank on every outlet / 100001 (above max)	ERR-03 inline message for the matching rule; Next stays disabled; previously entered valid values are preserved	Negative
TC-07	T07	Missing or invalid billing email	Outlet B's company has no billing email in the database	Enter valid amounts, click Next	ERR-04 naming the company. No quotation generated, no credits added	Negative
TC-08	T08	Submit valid top-up	Valid amounts entered, all billing emails valid	Click Next	Button disables, idempotency key created, quotation request sent to QuickBooks	Positive
TC-09	T09	QuickBooks unavailable or times out	Submission in progress; QuickBooks API returns error/timeout	Click Next	ERR-05 shown. No quotation exists, NO credits added to any company. Retry offered	Negative
TC-10	T10	Retry after QuickBooks failure	ERR-05 displayed; QuickBooks restored	Click Retry	Exactly ONE quotation is created (same idempotency key). No duplicate quotation, no double credit	Recovery
TC-11	T11	Quotation generated successfully	Valid submission, QuickBooks healthy	Click Next	Quotation created with one line item per company, amounts matching the entered top-ups	Positive
TC-12	T12	Credit write fails after quotation created	Quotation created; credit transaction fails mid-way	Complete submission	ERR-06 shown. Transaction rolled back so NO company is partially credited. Quotation voided or flagged and finance alerted	Negative
TC-13	T13	Credits committed per company	Quotation created successfully	Complete submission	Each company's balance increases by its own amount only. Balances of other companies under the same brand are unchanged	Positive
TC-14	T14	Quotation email fails to send	Credits committed; mail service down	Complete submission	WARN-01 shown. Credits REMAIN valid, quotation remains valid, Resend option offered and retry queued	Negative
TC-15	T15	Resend quotation email	WARN-01 displayed; mail service restored	Click Resend	Email delivered to the company's billing address. Credits are not applied a second time	Recovery
TC-16	T16	Full happy path to success screen	All services healthy	Complete the full flow	Quotation emailed, credits applied, audit log written, success screen shows quotation number and updated balance per company	Positive
TC-17	T17	Cancel or navigate away mid-flow	Amounts entered but Next not clicked	Navigate away / cancel	No quotation, no credits, no audit entry. Returning to the page starts a	Negative

					clean flow	
TC-18	T18	Double submission prevented	Valid amounts entered	Click Next twice rapidly	Only one quotation and one credit transaction. Button disabled after first click; idempotency key blocks the duplicate	Negative
TC-19	T19	Session expires during submission	Submission in progress, session token expires	Complete submission	User is asked to re-authenticate. After login, exactly one quotation and one set of credits exist	Negative
TC-20	T20	Concurrent top-up on the same company	Two finance executives top up the same company at the same time	Both submit 500	Final balance = original + 500 + 500. Neither write is lost (row-level lock / atomic increment)	Negative
TC-21	B1	Credits are company-scoped, not brand-scoped	Brand has outlets under Company X and Company Y	Top up only an outlet belonging to Company X	Only Company X's balance changes. Company Y is untouched. Quotation is billed to Company X only	Business rule
TC-22	B2	Multiple companies in one submission	Brand has outlets under Company X and Company Y	Top up 500 for X's outlet and 800 for Y's outlet	Two separate quotations or two clearly separated line items per billing entity, each emailed to its own company billing address; X +500, Y +800	Business rule
TC-23	B3	Partial fill of outlet list	Brand has 3 outlets	Fill only outlet A, leave B and C blank	Only outlet A's company is quoted and credited. B and C are excluded entirely from the quotation	Business rule
TC-24	B4	Quotation totals and rounding	Top-up values that produce a fractional tax amount	Submit	Line totals, tax and grand total on the quotation match the system calculation to 2 decimal places, with no rounding drift	Business rule
TC-25	B5	Balance is incremented, not overwritten	Company X has 2,000 existing credits	Top up 500	Balance becomes 2,500, not 500	Business rule
TC-26	B6	Audit trail completeness	Successful top-up	Complete the flow, then open the audit log	Entry records user, brand, each company, each amount, quotation number and timestamp	Business rule

5. Highest-risk areas

- **Money and credits out of step (TC-09, TC-10, TC-12, TC-18, TC-19).** The costly failure is a client being charged without credits, or credited twice. Idempotency and the atomic transaction are what prevent it.
- **Crediting the wrong entity (TC-21, TC-22, TC-23).** The brand-to-company split is the trickiest rule in the feature, so it needs its own tests.
- **Silent partial success (TC-12, TC-14).** The user must always be told exactly which of quotation, credits and email succeeded.